

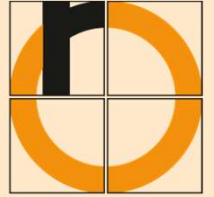


Systems Engineering - SySe

Chapter 6 – Design Thinking

Prof. Dr. Silke Lechner-Greite

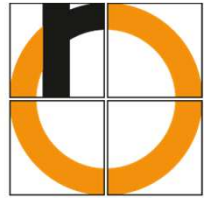
Content



Design Thinking and TRIZ

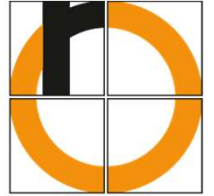
- What is Design Thinking?
- What is TRIZ?
- Design Thinking Workshop

Learning objectives

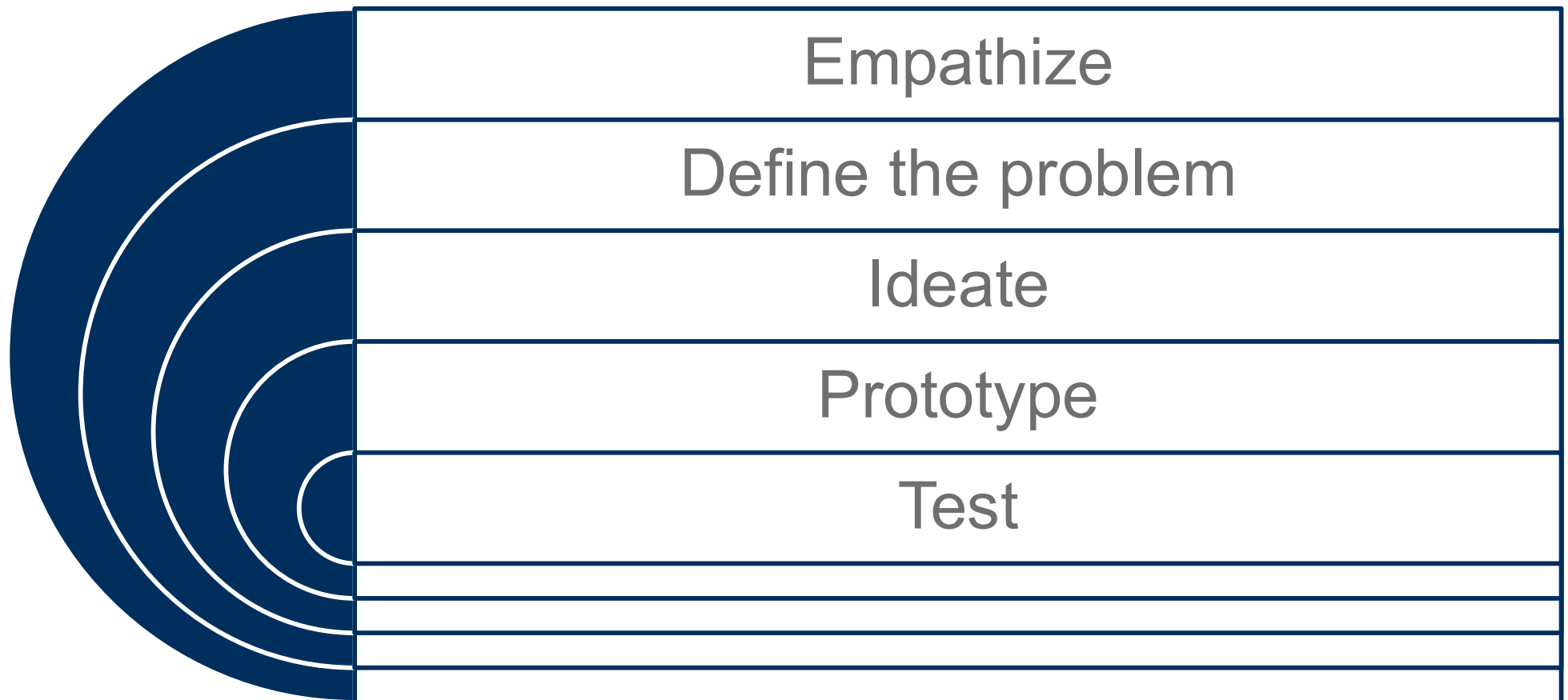


- Get an idea of other innovative approaches to finding solutions to a specific problem:
- Design Thinking - theory and work through of an online workshop on Design Thinking (d.school Stanford Starter Kit)
- TRIZ - learn more about TRIZ and its use to create innovative solutions.

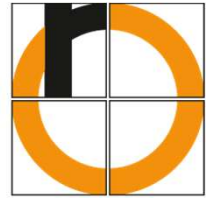
Design Thinking



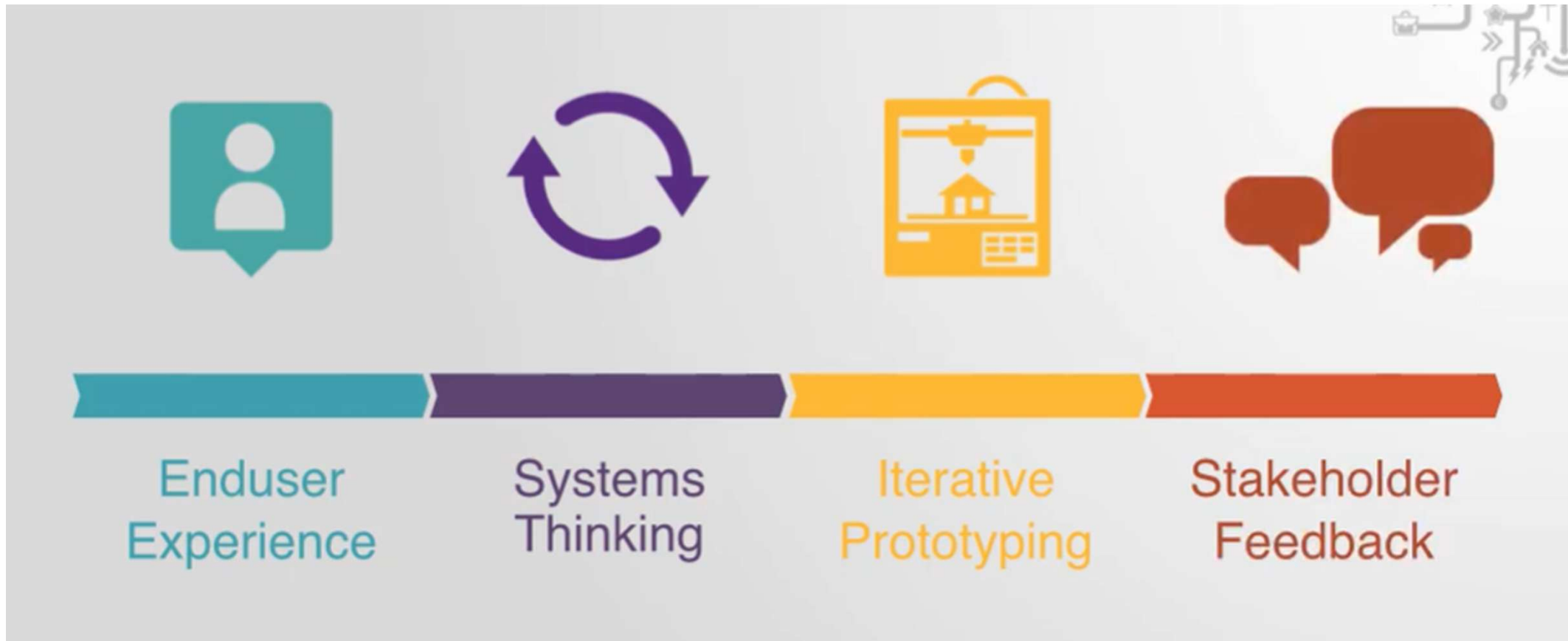
- What is design thinking? What are the key takeaways? Let's check on this short video listed on the following webpage (~4 min):
- <https://innovators-guide.ch/2013/02/design-thinking-2/>



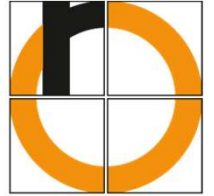
Design Thinking



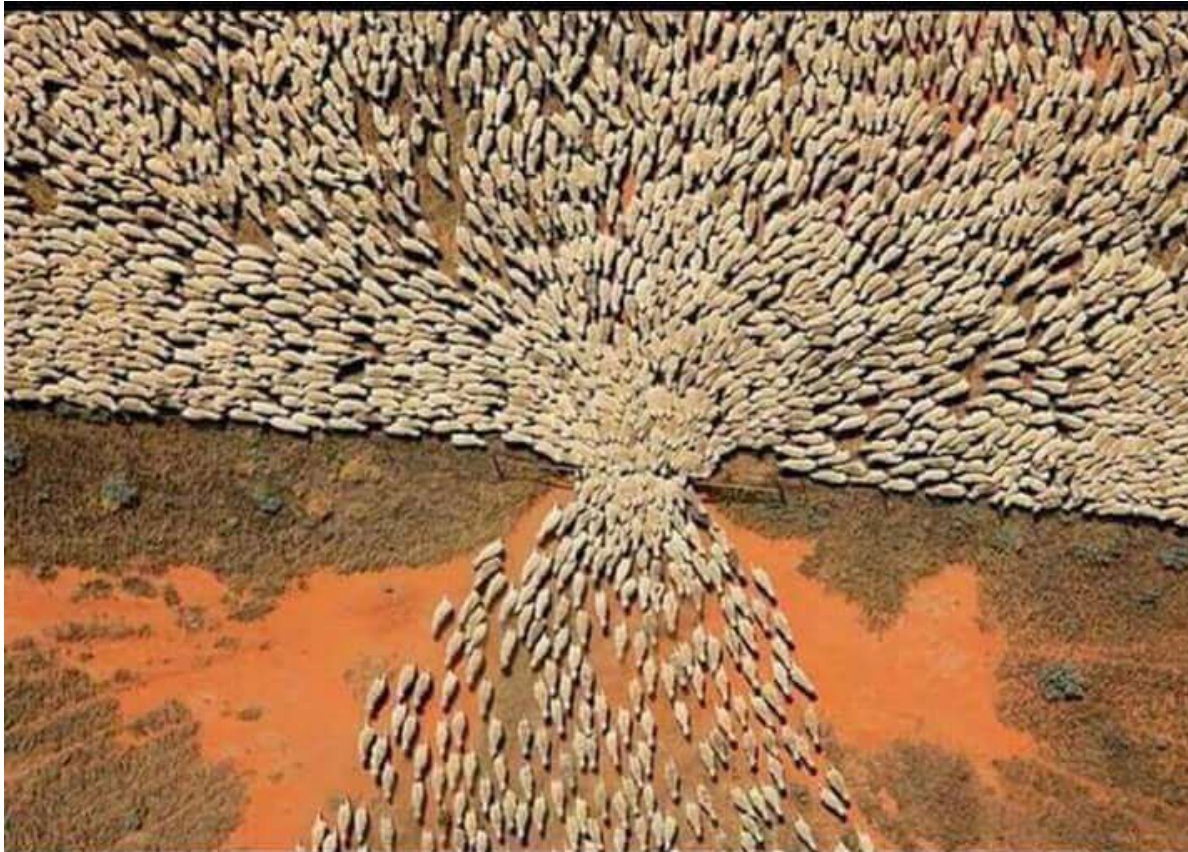
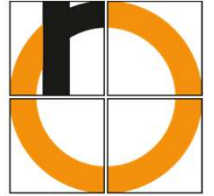
- What is Design Thinking in the context of complex technical systems (~ 8min):
<https://youtu.be/WrdSkqRypsg>
- Let's check on this video ...



Design Thinking



- Not enough?
- Check out these videos on design thinking
- Stanford Webinar on Design Thinking (~ 50 min):
<https://www.youtube.com/watch?v=vSuK2C89yjA>
- TEDx Talk: Professor Weinberg (Hasso Plattner Institut, Potsdam): Design Thinking (~1 h):
<https://www.youtube.com/watch?v=BePDcXCHvyA>



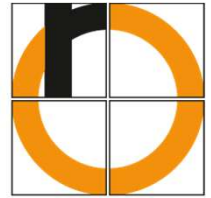
A herd of sheep is leaving the stall. There is no fence,
only the gate ...

"The Trap of thinking"

Picture: [pinterest.com](https://www.pinterest.com)

For years there was a fence.
Not anymore.
The sheep still have the fence in
their "head".
The brain prefers to work with
existing patterns - this is energy
saving.

How to develop innovative ideas?



Brainstorming: finding ideas

A group generates ideas for the solution of a specific problem:

- No evaluation of ideas allowed.
- Relax. Let images/ideas simply flow into your thoughts.
- Encourage wild, exaggerated ideas.
- Mass counts, not quality.
- Every person and every idea has equal rights
- Build on ideas from predecessors → associative

TRIZ: identification of existing problem

Define the existing system and identify all elements, interactions and available resources.

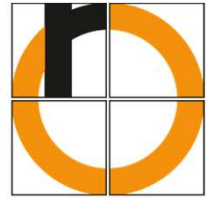
- Define the problem, resources & functionalities.
- Analyze the system and focus on understanding the problem.
- Is it a lack of understanding of the problem?
- Technical or physical contradiction?
- If I don't know how to solve it, focus on IP in and outside my field of technology

Design Thinking: change of perspective on the problem

„Design Thinking ist eine systematische Herangehensweise an komplexe Problemstellungen aus allen Lebensbereichen.“ (3)

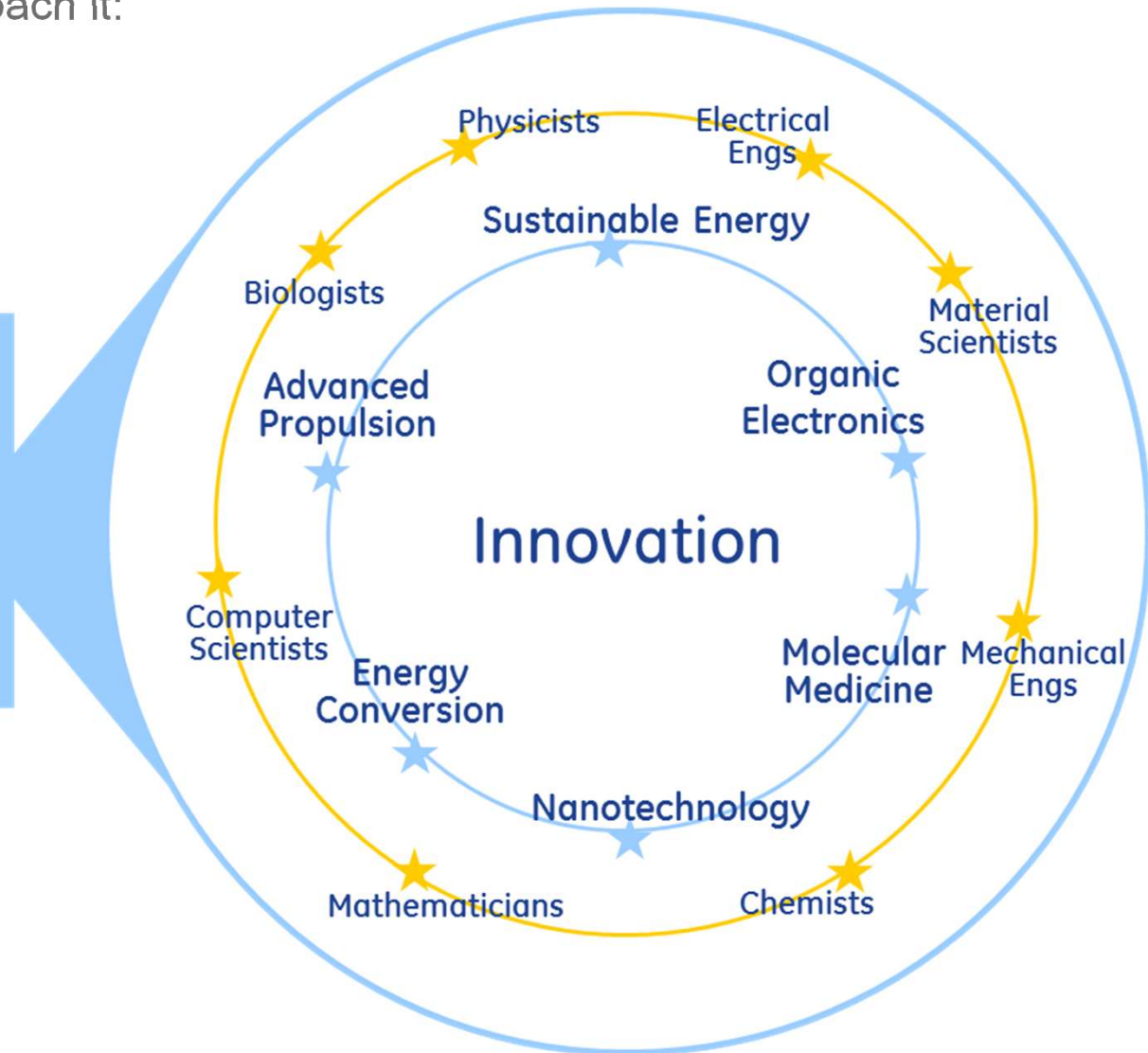
- Traditional approaches to approach the problem from a purely technical point of view should be overcome.
- Design Thinking promotes creativity and innovation.
- People are at the center of the idea generation process.

How to develop innovative ideas?

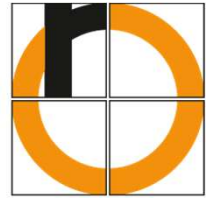


- That is how companies approach it:

**Innovation
occurs at the
intersection
of disciplines**



Design thinking happens in teams

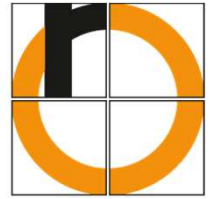


- Work in multidisciplinary teams
- Size of 5 to 6 people
- Common language
- “Blick über den Tellerrand”
- Bringing in several specialist skills
- No competitive thinking; it is about the “we”
- But: the team situation is probably the most conflicting area (performance-oriented teaching system, starting already at kindergarden)



Image taken from Hasso Plattner Institute
Webpage: <https://hpi.de/school-of-design-thinking/design-thinking/was-ist-design-thinking.html>

Ideas need a creative environment



<https://images.app.goo.gl/yrzHLV1G33ATnKJJ6>



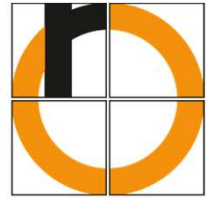
<https://images.app.goo.gl/VPBDjFTtMmgVYCsb8>



<https://medium.com/prodtype/how-to-utilize-a-design-thinking-workshop-389db64512e0>

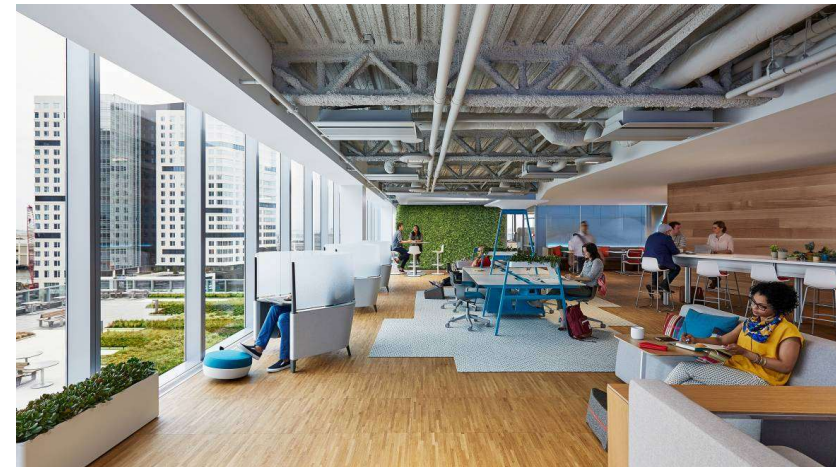
- So called creative rooms are utilized to encourage creativity
- Variable rooms adapt based on the project the team is working on.
- Furniture that can be moved around.
- Whiteboards, sticky notes, pens in various colors, flip charts, presentation areas, building blocks, prototyping materials, etc.
- Team supported environment – represent ideas visually, create haptical memory, share your information within teams → **variable room**
- <https://dschool.stanford.edu/>

Also, companies start to “inspire”



- Open Space offices – enhance interaction
- Huddle rooms
- Creativity rooms --- creative spaces

<https://www.steelcase.com/eu-de/raume-inspiration/unser-arbeitsplatz-designansatz/>

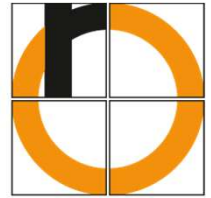


<https://www.diepresse.com/592229/kreativraume-gehen-wir-spielen>



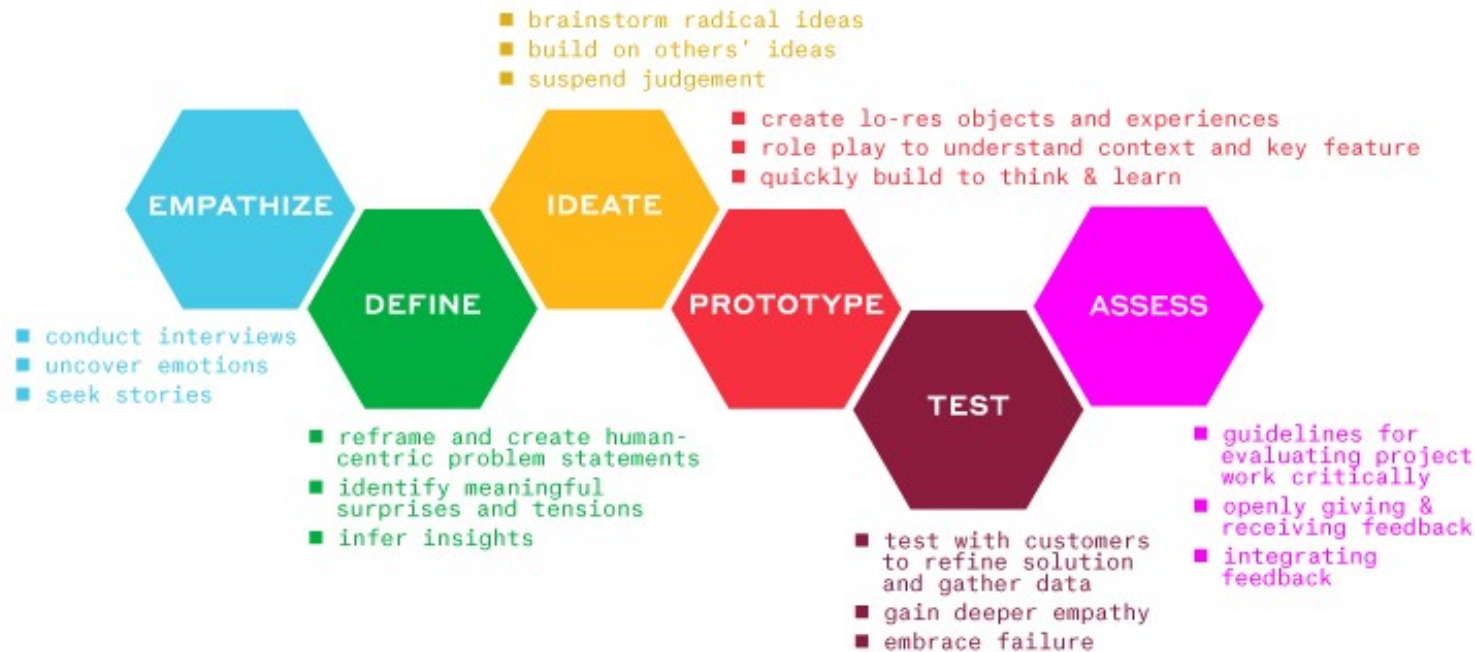
<https://www.neu-innovation.de/>

The Process of Design Thinking



- Iterative Process: understand – observe (go there and ask questions) – synthesis – ideation – prototyping / testing – implementing

Design Thinking Process Diagram*



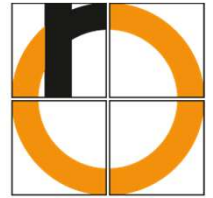
d.school Executive Education
Hasso Plattner Institute of Design at Stanford University

feedback loop by collecting feedback
through the prototype > evaluating >
redefining > ideate > refine prototype

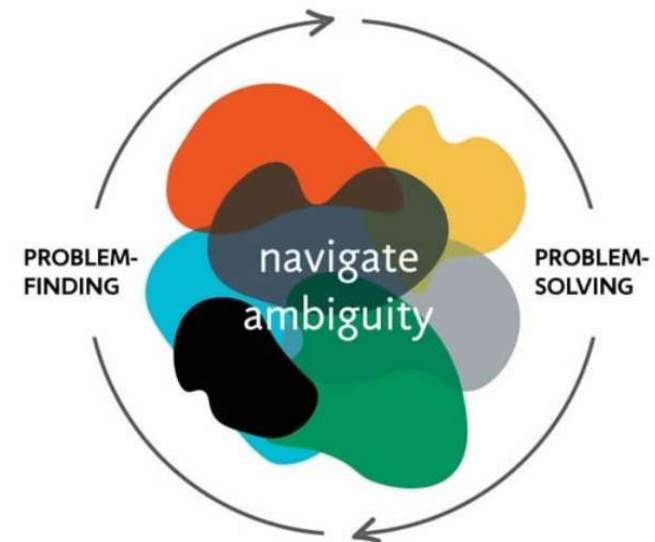
*not necessarily linear, apply as needed ©2019

- Primary focus: human aspect. Secondary focus: technology & business.

The 8 abilities of Stanford d.school



- <https://dschool.stanford.edu/about>
- Check out this website. What are the 8 abilities that make design thinking? What do they stand for? --- a bit further down the website
- Online Resources and guides on design thinking: <https://dschool.stanford.edu/resources>



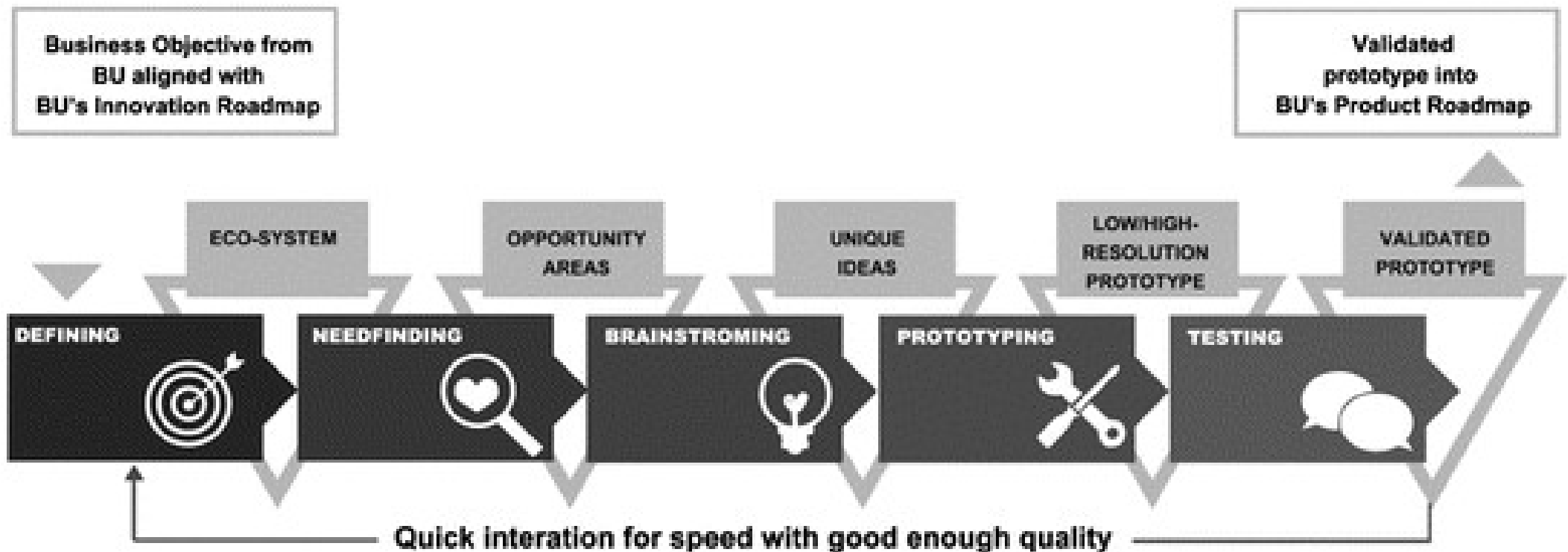
Picture: <https://dlibrary.stanford.edu/ambiguity>

Bootleg

- Design Thinking Bootleg is a set of tools and methods to be used for design thinking activities.
- <https://dschool.stanford.edu/resources/design-thinking-bootleg>
- Which method / tool do you find particularly interesting?
- Success stories for Design thinking: <https://theaccidentaldesignthinker.com/2017/09/16/40-design-thinking-success-stories/>
 - Browse through the examples
 - Choose one of them and summarize the core innovations
 - Why did you choose this example?

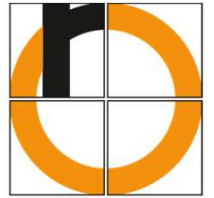
Industrial Design Thinking

- ... on the example of Industrial Design Thinking at Siemens Corporate Technology, China

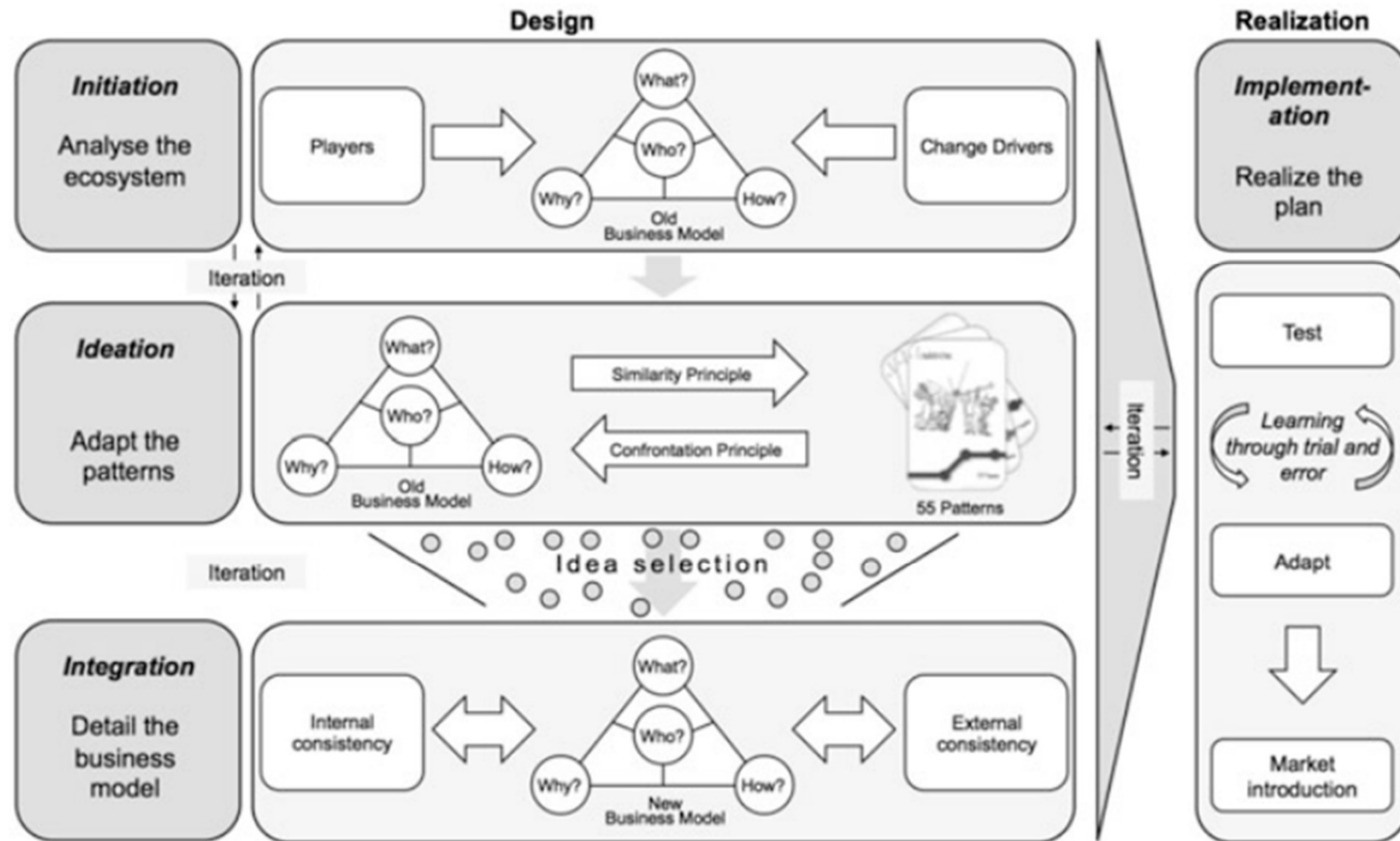


Source: Ge, X., Maisch, B. (2016). Industrial Design Thinking at Siemens Corporate Technology, China. In: Brenner, W., Uebernickel, F. (eds) Design Thinking for Innovation. Springer, Cham. https://doi.org/10.1007/978-3-319-26100-3_12

Integrate Design Thinking into Business Model Design processes



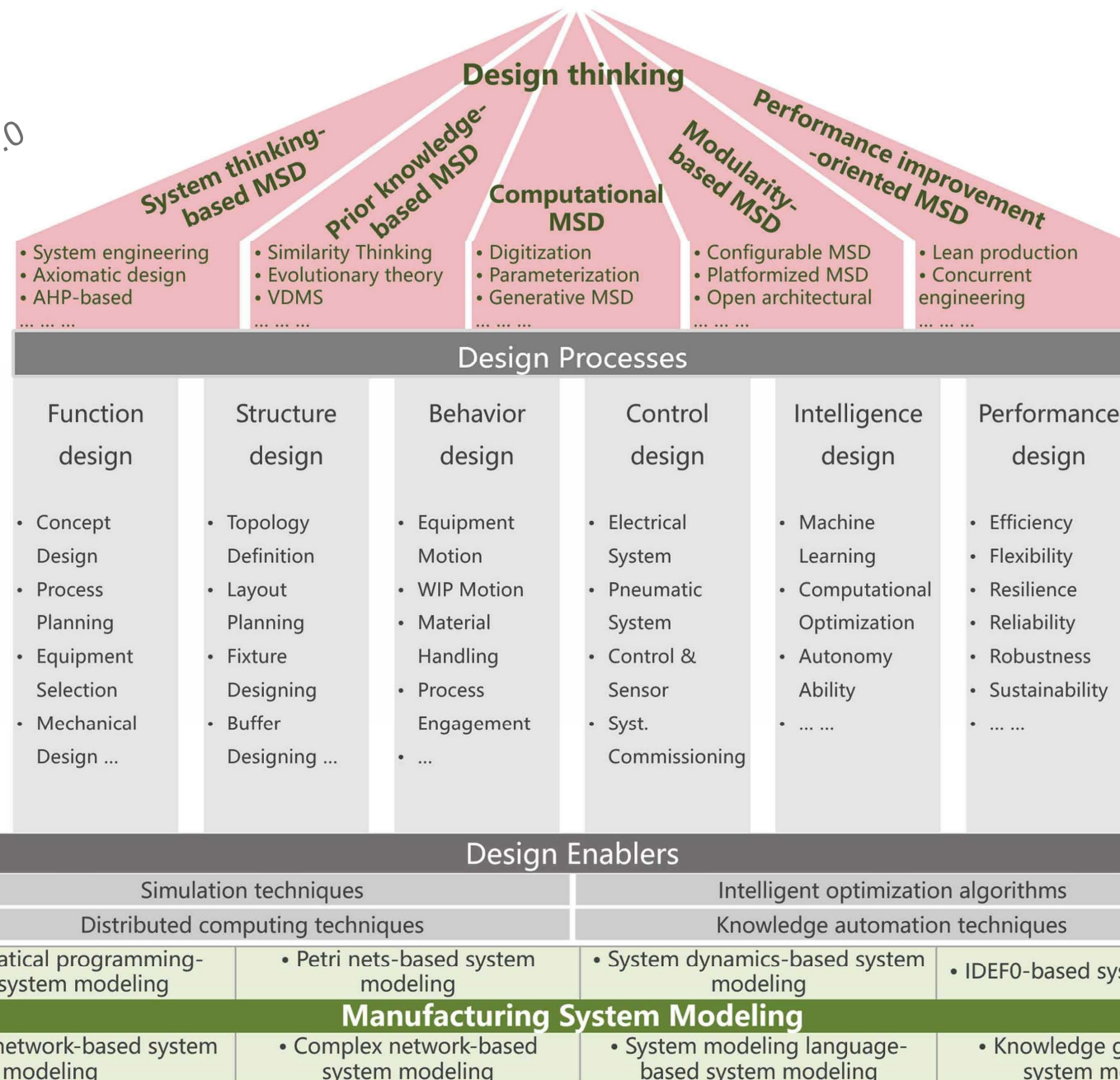
- Core question: how to introduce Design Thinking into a companies business model?



Source: Ge, X., Maisch, B. (2016). Industrial Design Thinking at Siemens Corporate Technology, China. In: Brenner, W., Uebernickel, F. (eds) Design Thinking for Innovation. Springer, Cham. https://doi.org/10.1007/978-3-319-26100-3_12

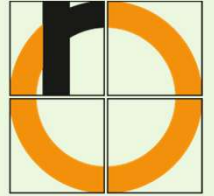
Manufacturing system design (MSD)

Keywords:
Industry 4.0 and Industry 5.0



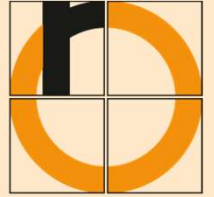
Source: Leng et al.,
Review of
manufacturing system
design in the interplay of
Industry 4.0 and
Industry 5.0 (Part I):
Design thinking and
modeling methods,
Journal of
Manufacturing Systems,
Volume 76, **2024**,
Pages 158-187, ISSN
0278-6125,
<https://doi.org/10.1016/j.jmsy.2024.07.012>.

d.School starter kit



- This is what we'll work through within the lecture:
 - Starter Kit on Design Thinking: <https://dschool.stanford.edu/resources/dschool-starter-kit>
 - Online Course: https://docs.google.com/presentation/d/1kMROhf-S6z0hLKb7Km0PddUff-wiYkuD5QCNeD8t-Ns/edit#slide=id.g827ccf4a3e_0_8

TRIZ



- Please review the slide courtesy of Oliver Mayer on LC.